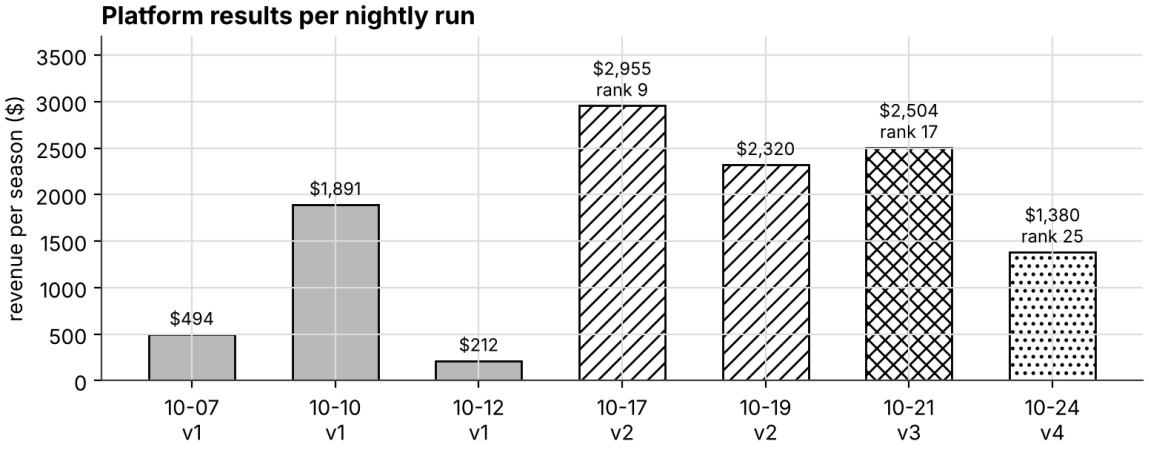
Dynamic Pricing Competition – Algorithm Summary

BenevolentSwan (Ali) · Learning Optimization, BHT Berlin · status 27 Oct 2026 · short version for discussion

Setup: 2 airlines, 80 seats, 100 days per season, 100 seasons per simulation. The demand model is random per simulation (Poisson arrivals, demand rising toward departure, random level, random willingness to pay, random sensitivity to the price gap). Goal: maximise revenue.

1. Versions at a glance

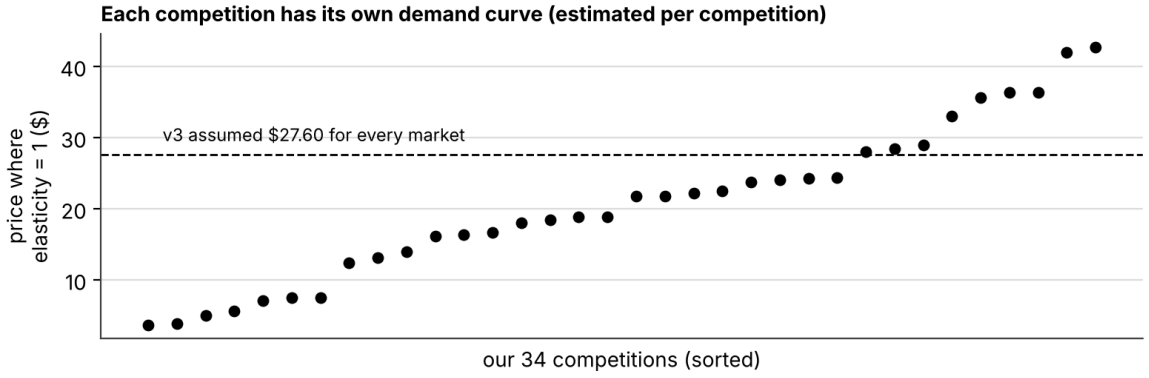
Version	How it prices	Real result	Worked	Did not work
v1 copycat	copies competitor's last price	$212 – $1,891 /season	fine vs a sensible opponent	opponent decides everything; ignores seats, time, demand
v2 booking curve	target seats-sold curve (top 20 % seasons); behind → undercut, ahead → mark up; fixed monopoly ($55/68/78) and clearance ($14–22) prices	**$2,955, rank 9** (Oct 17); $2,320 (Oct 19)	pacing: 8 of 12 competitions sold out at $2,830–4,114	fixed prices ignore days left (22–32 seats empty vs GorgeousNewt); follows expensive rival in "cheap" markets (nobody buys above ~$40)
v3 fluid model	Poisson demand model from past data; each day: price = argmax price × min(seats left, predicted sales)	$2,504, rank 17 (Oct 21)	best competitions so far ($4,381, $4,141)	one fixed model for all markets → stuck at $27.50 vs RationalLion ($555, $624)
v4 online learning	v3 + learns market size a and price sensitivity b per competition (Newton's method, prior)	$1,380, rank 25 (Oct 24, all vs KickassMoth)	learned the market (prices $44–66, up to $3,900/season)	fixed "cheaper bonus" + assumed rival price fixed → price war after season 51
v5 reaction-aware	v4 + learns cheaper bonus + detects a rival that follows my price, then picks the best level	pending (Oct 26)	simulator with reacting opponents: $2,270 vs v4 $2,133 vs v2 $1,927 (−2 % vs opponents that ignore me)	
OLS variant	course Week-3 method: OLS per 20-day window, learned online, grid search	not uploaded	simulator: $2,262 vs v5 $2,472 (v5 better in 18/25 markets)	

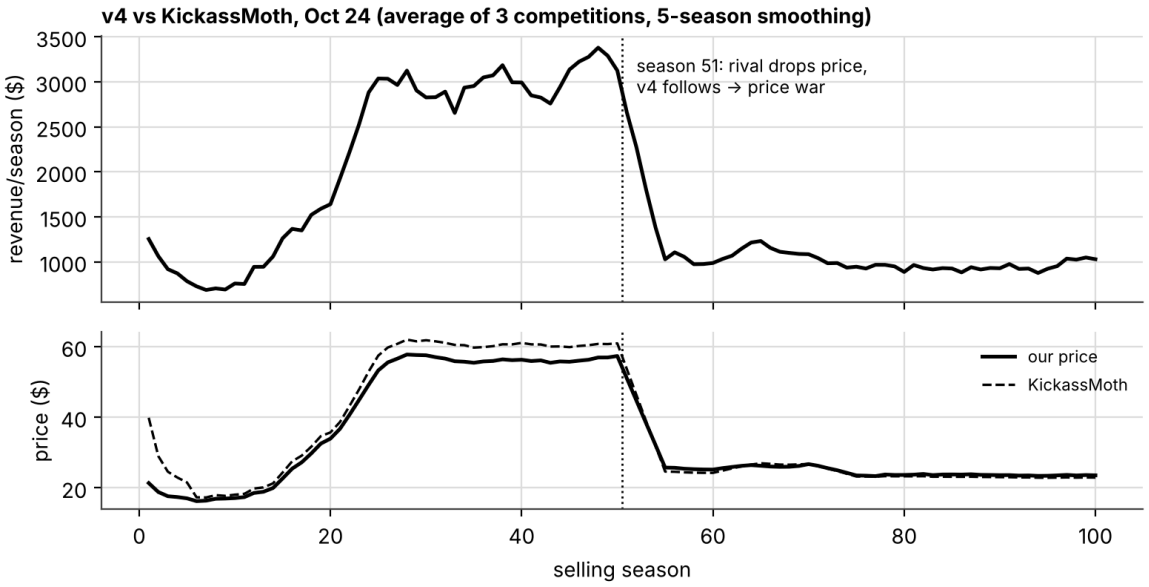


2. Key findings from the data

1. **Every competition has its own demand curve.** The price where elasticity = 1 ranges from $3.90 to $42.70 across 34 competitions → one pooled model fails (v3); learn per competition (v4).
2. **Capacity rule:** best price = max(price with elasticity = 1, run-out price for the remaining days). Fixed monopoly/clearance prices break it (v2 vs GorgeousNewt).
3. **Competitors react.** Oct 24: rival price today vs my price yesterday correlation 0.91 (and reverse). Sales depended on being cheaper (0.5–0.6/day) vs dearer (0.3/day) at any level from $15 to $70. Treating the rival price as fixed led to a price war (v4).

4. **Exploration is needed:** copying the competitor makes the two prices collinear, so regression cannot estimate the own-price effect.
5. **Testing:** my first simulator missed extreme markets, so v3 looked safe. The simulator must include the cases that break an algorithm.





3. Methods used (course link)

Method	Where
price matching / inventory monitoring / booking curve	v1, v2
elasticity, revenue optimum, capacity rule (Phillips ch. 3)	v3–v5 price choice
demand estimation by regression (course: OLS; mine: Poisson)	v3 (pooled), v4/v5 (per competition), OLS variant
Newton's method (Oct 20)	v4/v5 parameter fit
grid search (derivative-free)	price choice (revenue has kinks: min(), cheaper bonus)
simple game-theoretic response (leader/follower)	v5

4. Open questions

- Poisson regression instead of OLS acceptable if both are compared on test seasons?
- Newton with a prior vs scipy.optimize (BFGS / trust region) for the fit?
- Rival model: leader/follower (v5), ARIMAX forecast, or Nash equilibrium?
- Next: learn the time pattern per competition; dynamic programming over (day, seats left) with a table loaded from an uploaded file; tune remaining parameters by optimisation in the simulator.